

Financial Mathematics Practice Worksheet

Chapter 11.6 – Mortgage Repayments

Tasks 1–13, sorted from easiest to hardest (difficulty 1/5 to 5/5, with Tasks 11–13 as extra harder capstone comparisons). Key ideas: a loan of K dollars repaid in n equal instalments of a dollars per period, at periodic interest rate r , satisfies $K = (a/r) \cdot [1 - (1+r)^{-n}]$, so the required payment is $a = rK/[1 - (1+r)^{-n}]$. Each payment splits into an interest part (r times the outstanding balance from the previous period) and a principal part (the rest of the payment, which reduces the balance). If the first payment is made immediately rather than at the end of the first period (an annuity due), the first payment counts at full face value and only the remaining $n-1$ payments are discounted. When the payment amount a is fixed in advance instead of solved for, the number of periods needed is the smallest integer n with $n \geq [\ln a - \ln(a-rK)]/\ln(1+r)$; unless this holds with equality, the final payment will be smaller than a . Finally, when several deposits are made within a single interest period and the bank applies simple interest to money held for only part of that period, each deposit is multiplied by $(1+r \cdot t)$, where t is the remaining fraction of the period after the deposit is made. Tasks 11–13 additionally compare the present value of several complete payment schedules against each other — including schedules that combine an upfront cash portion with a later series of equal payments — to determine which schedule is cheapest, and how that conclusion can change at a different interest rate. Round dollar amounts to 2 decimal places unless otherwise noted.

Task 1. Equipment Loan: Finding the Equal Annual Payment

Difficulty: 1/5

A small bakery-supply distributor borrows \$60,000 at the beginning of the year to buy a delivery van and a walk-in cooler. The loan is to be repaid in 6 equal instalments at the end of each year, with interest charged at 12% per year, compounding annually.

Statements (True / False):

- a) The periodic interest rate to use in the payment formula is $r = 0.12$.
- b) The required equal annual payment is approximately \$14,593.54.
- c) The interest portion of the first payment is exactly \$7,200.00.
- d) The principal-repayment portion of the first payment is more than half of the total payment amount.
- e) The outstanding balance immediately after the second payment is approximately \$45,000.00.

Solution – Task 1

Given:

- K (loan amount) = \$60,000
- r (annual interest rate) = 0.12
- n = 6 equal annual payments

Formula(s):

- Equal-payment formula: $K = (a/r) \cdot [1 - (1+r)^{-n}]$, so $a = rK/[1 - (1+r)^{-n}]$
- Interest in a period = $r \times$ outstanding balance at the start of that period
- Principal repaid in a period = payment – interest for that period; new balance = old balance – principal repaid

Step-by-Step Calculations:

$$a = 0.12 \times 60,000 / [1 - (1.12)^{-6}] = 7,200 / [1 - 0.506631] = 7,200 / 0.493369 \approx \$14,593.54.$$

$$\text{Year-1 interest} = 0.12 \times 60,000 = \$7,200.00.$$

Year-1 principal repaid = $14,593.54 - 7,200.00 = \$7,393.54$; half of the payment is $14,593.54/2 = \$7,296.77$, and $7,393.54 > 7,296.77$.

$$\text{Balance after year 1} = 60,000 - 7,393.54 = \$52,606.46.$$

$$\text{Year-2 interest} = 0.12 \times 52,606.46 = \$6,312.77; \text{ Year-2 principal repaid} = 14,593.54 - 6,312.77 = \$8,280.77.$$

$$\text{Balance after year 2} = 52,606.46 - 8,280.77 = \$44,325.69.$$

Explanation of Each Statement:

- a) TRUE.** A 12% annual interest rate charged once per year, compounding annually, is simply used as the decimal $r = 0.12$ in every calculation for this loan.
- b) TRUE.** Solving $a = rK/[1 - (1+r)^{-n}]$ with $r=0.12$, $K=60,000$, $n=6$ gives $a \approx \$14,593.54$, matching exactly.
- c) TRUE.** In the very first period, the entire \$60,000 principal has been outstanding for the full year, so the interest charged is simply 12% of \$60,000, which is exactly \$7,200.00 with no rounding needed.
- d) TRUE.** Subtracting the \$7,200.00 interest charge from the \$14,593.54 payment leaves \$7,393.54 for principal reduction. Comparing that to half of the payment (\$7,296.77) shows the principal portion is indeed slightly larger than half, even in the very first year when interest is at its highest relative to later years.
- e) FALSE.** Carrying the amortization forward two full years — recomputing interest on the shrinking balance each time and subtracting the resulting principal portion — gives a balance of approximately \$44,325.69 after the second payment, not \$45,000.00.

Final Answers — a: TRUE b: TRUE c: TRUE d: TRUE e: FALSE

Task 2. Auto Dealership Financing with Monthly Compounding

Difficulty: 1.5/5

A used-car dealership finances a customer's \$24,000 vehicle purchase over 4 years, with equal payments due at the end of each month and interest charged at a nominal annual rate of 9%, compounding monthly.

Statements (True / False):

- a) The monthly periodic interest rate used in the payment formula is $r = 0.75\%$.
- b) The required monthly payment is approximately \$597.24.
- c) The total amount paid over all 48 monthly payments is approximately \$29,500.00.
- d) The total interest paid over the life of the loan is approximately \$4,667.57.
- e) If the same \$24,000 were instead repaid in 4 equal annual instalments at the same nominal 9% rate, the required annual payment would be less than \$2,388.96.

Solution – Task 2

Given:

- K (amount financed) = \$24,000
- Nominal annual rate = 9%, compounding monthly
- Term = 4 years (48 monthly payments)

Formula(s):

- Monthly periodic rate: $r = \text{nominal annual rate} / 12$
- Equal-payment formula: $a = rK/[1-(1+r)^{-n}]$
- Total paid = $a \times n$; total interest = total paid – K

Step-by-Step Calculations:

$$r = 0.09/12 = 0.0075; n = 4 \times 12 = 48 \text{ months.}$$

$$a = 0.0075 \times 24,000 / [1 - (1.0075)^{-48}] = 180 / 0.301329 \approx \$597.24.$$

$$\text{Total paid} = 597.24 \times 48 \approx \$28,667.57 \text{ (not } \$29,500.00).$$

$$\text{Total interest} = 28,667.57 - 24,000.00 \approx \$4,667.57.$$

$$\text{For 4 equal annual instalments at } r=0.09, n=4: a = 0.09 \times 24,000 / [1 - (1.09)^{-4}] = 2,160 / 0.291676 \approx \$7,408.05, \text{ which is far more than } \$2,388.96, \text{ not less.}$$

Explanation of Each Statement:

- a) **TRUE.** Dividing the 9% nominal annual rate by 12 monthly periods gives a periodic rate of 0.75%, which is what the payment formula requires when payments and compounding both happen monthly.
- b) **TRUE.** Plugging $r=0.0075$, $K=24,000$, $n=48$ into the payment formula gives $a \approx \$597.24$, matching exactly.
- c) **FALSE.** Multiplying the correct monthly payment by 48 months gives a total of approximately \$28,667.57, not \$29,500.00 — the stated figure overstates the true total by a few hundred dollars.
- d) **TRUE.** Subtracting the \$24,000 principal from the correctly computed total paid of about \$28,667.57 leaves approximately \$4,667.57 in total interest.

e) FALSE. Spreading the same debt over only 4 much larger annual payments, rather than 48 smaller monthly ones, actually requires a much bigger payment per period — about \$7,408.05 — which is far above \$2,388.96, not below it; fewer, larger periods mean each payment carries proportionally more principal and interest packed into it.

Final Answers — a: TRUE b: TRUE c: FALSE d: TRUE e: FALSE

Task 3. Restaurant Renovation Loan: Tracking the Amortization Schedule

Difficulty: 2/5

A restaurant owner borrows \$45,000 to renovate the dining room, agreeing to repay the loan in 5 equal annual instalments at the end of each year, with interest at 10% per year compounding annually.

Statements (True / False):

- a) The required equal annual payment is approximately \$11,870.89.
- b) The interest portion of the first payment is \$5,000.00.
- c) The outstanding balance immediately after the third payment is approximately \$20,602.37.
- d) In the fourth year, the interest portion of the payment is larger than the principal portion of the payment.
- e) Adding up the principal-repayment portions of all 5 payments gives a total of exactly \$46,200.00.

Solution – Task 3

Given:

- K (loan amount) = \$45,000
- r (annual interest rate) = 0.10
- n = 5 equal annual payments

Formula(s):

- Equal-payment formula: $a = rK/[1-(1+r)^{-n}]$
- Interest in a period = $r \times$ outstanding balance at the start of that period
- Principal repaid = payment – interest; new balance = old balance – principal repaid

Step-by-Step Calculations:

$$a = 0.10 \times 45,000 / [1 - (1.10)^{-5}] = 4,500 / 0.379079 \approx \$11,870.89.$$

Year-1: interest = $0.10 \times 45,000 = \$4,500.00$; principal = $11,870.89 - 4,500.00 = \$7,370.89$; balance = $\$37,629.11$.

Year-2: interest = $0.10 \times 37,629.11 = \$3,762.91$; principal = $\$8,107.98$; balance = $\$29,521.14$.

Year-3: interest = $0.10 \times 29,521.14 = \$2,952.11$; principal = $\$8,918.77$; balance = $\$20,602.37$.

Year-4: interest = $0.10 \times 20,602.37 = \$2,060.24$; principal = $11,870.89 - 2,060.24 = \$9,810.65$; here interest (\$2,060.24) is much smaller than principal (\$9,810.65).

Year-5: interest = $0.10 \times 10,791.72 = \$1,079.17$; principal = $\$10,791.72$; balance = $\$0.00$. Summing all five principal portions:

$$7,370.89 + 8,107.98 + 8,918.77 + 9,810.65 + 10,791.72 = \$45,000.00 \text{ exactly.}$$

Explanation of Each Statement:

- a) **TRUE.** Solving $a = rK/[1-(1+r)^{-n}]$ with $r=0.10$, $K=45,000$, $n=5$ gives $a \approx \$11,870.89$, matching exactly.
- b) **FALSE.** In the first year the full \$45,000 principal is outstanding, so the interest charged is 10% of \$45,000, which is \$4,500.00, not \$5,000.00.

- c) **TRUE**. Carrying the amortization schedule forward year by year — recomputing interest on the shrinking balance each time — gives an outstanding balance of approximately \$20,602.37 immediately after the third payment.
- d) **FALSE**. By the fourth year the balance has already shrunk substantially, so the interest portion of that payment (about \$2,060.24) is much smaller than the principal portion (about \$9,810.65) — the opposite of what the statement claims. As a loan amortizes, later payments always shift further toward principal and away from interest.
- e) **FALSE**. Since the loan is fully paid off by the end of year 5 (the balance reaches exactly \$0.00), the principal portions of all five payments must add up to precisely the original \$45,000.00 borrowed — not \$46,200.00.

Final Answers — a: TRUE b: FALSE c: TRUE d: FALSE e: FALSE

Task 4. Franchise Buy-In Paid as an Annuity Due

Difficulty: 2.5/5

A new franchisee agrees to pay \$150,000 for the rights to open a location, structured as 10 equal annual payments, with the first payment due immediately and each following payment due at the beginning of each subsequent year. The agreed annual interest rate is 11%.

Statements (True / False):

- a) The combined present-value factor multiplying the payment is approximately 6.537048.
- b) The required equal payment a is approximately \$22,946.14.
- c) If instead the first payment were due at the end of year 1, the required equal payment would be lower than \$22,946.14.
- d) The difference between the ordinary-annuity payment and the annuity-due payment is approximately \$2,524.08.
- e) The total of all 10 annuity-due payments combined is approximately \$220,000.00.

Solution – Task 4

Given:

- K (present value owed) = \$150,000
- r (annual interest rate) = 0.11
- $n = 10$ total payments, first one immediate

Formula(s):

- Annuity-due present value: $K = a + (a/r) \cdot [1 - (1+r)^{-(n-1)}]$
- Ordinary-annuity comparison: $a_{\text{ordinary}} = rK/[1 - (1+r)^{-n}]$

Step-by-Step Calculations:

Combined factor = $1 + (1/0.11) \cdot [1 - (1.11)^{-9}] = 1 + 9.090909 \times (1 - 0.391110) = 1 + 5.537048 \approx 6.537048$.

$a = 150,000 / 6.537048 \approx \$22,946.14$.

Ordinary-annuity payment: $a_{\text{ordinary}} = 0.11 \times 150,000 / [1 - (1.11)^{-10}] = 16,500 / 0.647848 \approx \$25,470.21$, which is higher than \$22,946.14, not lower.

Difference = $25,470.21 - 22,946.14 \approx \$2,524.08$.

Total of all 10 due payments = $22,946.14 \times 10 \approx \$229,461.39$, not \$220,000.00.

Explanation of Each Statement:

- a) **TRUE**. Because the very first payment happens at time zero, it isn't discounted at all and simply counts as a itself; the remaining 9 payments form an ordinary annuity discounted back from year 1, giving the combined multiplier of approximately 6.537048 shown above.
- b) **TRUE**. Dividing the \$150,000 owed by the combined factor of 6.537048 gives $a \approx \$22,946.14$, matching exactly.
- c) **FALSE**. Delaying the very first payment to the end of year 1 instead of paying it immediately means the lender receives money later and thus needs a larger payment to compensate — the ordinary-annuity payment of about

\$25,470.21 is actually higher than the due-annuity payment, not lower.

d) TRUE. Subtracting the due-annuity payment from the ordinary-annuity payment gives $25,470.21 - 22,946.14 \approx \$2,524.08$, matching exactly.

e) FALSE. Multiplying the correct per-payment amount of \$22,946.14 by the 10 total payments gives approximately \$229,461.39 in total, not \$220,000.00.

Final Answers — a: TRUE b: TRUE c: FALSE d: TRUE e: FALSE

Task 5. Fixed Annual Payment on a Working-Capital Loan

Difficulty: 3/5

A landscaping company borrows \$35,000 at 13% annual interest and commits to repaying exactly \$10,000 at the end of each year (a fixed amount chosen by the company, not solved from the equal-payment formula), continuing until the debt is fully retired, with a smaller final payment to settle whatever remains.

Statements (True / False):

- a) The smallest whole number of yearly payments needed to pay off the loan is $n = 5$.
- b) The first four payments are each exactly \$10,000.00, totalling \$40,000.00.
- c) The fifth and final payment is approximately \$9,682.53.
- d) The total amount paid over the entire life of the loan is approximately \$49,682.53.
- e) The total interest paid over the life of the loan is less than the original \$35,000 principal.

Solution – Task 5

Given:

- K (loan amount) = \$35,000
- r (annual interest rate) = 0.13
- a (fixed annual payment) = \$10,000

Formula(s):

- Number of periods needed: smallest integer n with $n \geq [\ln a - \ln(a-rK)]/\ln(1+r)$; the final payment is smaller than a unless this holds exactly
- Future value of loan after m years: $K \cdot (1+r)^m$
- Future value of m fixed payments (valued at the time of the m -th payment): $(a/r) \cdot [(1+r)^m - 1]$

Step-by-Step Calculations:

$[\ln(10,000) - \ln(10,000 - 0.13 \times 35,000)]/\ln(1.13) = [\ln(10,000) - \ln(5,450)]/\ln(1.13) \approx 4.9663$, so the smallest integer n satisfying the condition is $n = 5$.

Four full payments of \$10,000 total $4 \times 10,000 = \$40,000.00$.

Future value of the loan after 4 years = $35,000 \times (1.13)^4 \approx \$57,066.58$.

Future value of the 4 payments of \$10,000, valued at year 4 = $(10,000/0.13) \times [(1.13)^4 - 1] \approx \$48,497.97$.

Remaining debt at year 4 = $57,066.58 - 48,497.97 \approx \$8,568.61$; carrying this forward one more year with interest gives the final payment: $8,568.61 \times 1.13 \approx \$9,682.53$.

Total paid = $40,000.00 + 9,682.53 \approx \$49,682.53$; total interest = $49,682.53 - 35,000.00 \approx \$14,682.53$, which is indeed less than \$35,000.00.

Explanation of Each Statement:

- a) **TRUE**. Evaluating $[\ln a - \ln(a-rK)]/\ln(1+r)$ with $a=10,000$, $r=0.13$, $K=35,000$ gives approximately 4.9663; since this is not a whole number, the smallest integer at or above it is 5, meaning 5 payments are needed.

- b) TRUE.** Since 4 full periods pass before the loan is paid off, the first four payments are each the full fixed amount of \$10,000.00, adding up to exactly \$40,000.00.
- c) TRUE.** Tracking the loan's growth at 13% for 4 years and comparing it to the accumulated value of the four \$10,000 payments over the same span shows a small remaining debt at that point; carrying that remainder forward one more year at 13% interest gives a final payment of approximately \$9,682.53.
- d) TRUE.** Adding the \$40,000.00 from the four full payments to the approximately \$9,682.53 final payment gives a total of about \$49,682.53 paid over the life of the loan.
- e) TRUE.** Subtracting the original \$35,000 principal from the total amount paid (about \$49,682.53) leaves roughly \$14,682.53 in total interest, which is indeed smaller than the \$35,000 principal itself.

Final Answers — a: TRUE b: TRUE c: TRUE d: TRUE e: TRUE

Task 6. Comparing Cash vs. Instalment Purchase of a Delivery Fleet

Difficulty: 3.5/5

A logistics company needs to acquire a fleet of delivery vehicles and is choosing between two payment schedules. Option A: pay \$500,000 in cash today. Option B: pay \$100,000 per year for 7 years, with the first payment due immediately and each subsequent payment due at the beginning of the following years. The company wants to know which option is cheaper in present-value terms, first at a 10% annual interest rate and then at a 14% annual interest rate.

Statements (True / False):

- a) At a 10% annual rate, the present value of Option B is approximately \$535,526.07.
- b) At a 10% annual rate, Option A is the cheaper choice.
- c) At a 14% annual rate, the present value of Option B is approximately \$495,000.00.
- d) At a 14% annual rate, Option B becomes the cheaper choice.
- e) Option B must always be the more expensive choice in present-value terms, no matter what the interest rate is.

Solution – Task 6

Given:

- Option A: \$500,000 paid today
- Option B: $a = \$100,000/\text{year}$, $n = 7$ payments, first one immediate
- Interest rates considered: 10% and 14% per year

Formula(s):

- Annuity-due present value: $PV = a + (a/r) \cdot [1 - (1+r)^{-(n-1)}]$

Step-by-Step Calculations:

At $r=0.10$: $PV_B = 100,000 + (100,000/0.10) \times [1 - (1.10)^{-6}] = 100,000 + 1,000,000 \times 0.435526 \approx \$535,526.07$.

Comparing: \$535,526.07 (Option B) > \$500,000.00 (Option A), so at 10% cash is cheaper.

At $r=0.14$: $PV_B = 100,000 + (100,000/0.14) \times [1 - (1.14)^{-6}] = 100,000 + 714,285.71 \times 0.544813 \approx \$488,866.75$ (not \$495,000.00).

Comparing: \$488,866.75 (Option B) < \$500,000.00 (Option A), so at 14% the instalment plan becomes cheaper.

Explanation of Each Statement:

- a) **TRUE.** Applying the annuity-due present-value formula with $a=100,000$, $r=0.10$, $n=7$ gives $PV_B \approx \$535,526.07$, matching exactly.
- b) **TRUE.** Since the present value of Option B at 10% (about \$535,526.07) is higher than the \$500,000 cash price, paying cash upfront is the cheaper option at this interest rate.

c) FALSE. Recomputing the annuity-due present value at the higher rate of 14% gives $PV_B \approx \$488,866.75$, not \$495,000.00.

d) TRUE. At 14%, Option B's present value (about \$488,866.75) drops below the \$500,000 cash price, meaning the instalment plan is now the cheaper option — a higher discount rate reduces the present value of future payments more than it affects the already-immediate cash price.

e) FALSE. The comparison in part (d) directly contradicts this claim: at 14% the instalment option, despite its larger total nominal payout, actually costs less in present-value terms than the cash option. Raising the interest rate shrinks the present value of payments spread out over time, so which option is cheaper genuinely depends on the interest rate used, not just on the raw total dollars paid.

Final Answers — a: TRUE b: TRUE c: FALSE d: TRUE e: FALSE

Task 7. Quarterly Savings Deposits with Annual Compounding

Difficulty: 4/5

At the end of each quarter, a small business owner deposits \$250 into a savings account that pays interest once per year, at an annual rate of 8%. Because each quarterly deposit sits in the account for only part of the year before the annual interest is credited, the bank applies simple interest to each deposit for the fraction of the year it remains on deposit before the year-end crediting date.

Statements (True / False):

- a) The year-end equivalent value of the four quarterly deposits made during a single year is \$1,100.00.
- b) The account balance after 4 years is approximately \$4,700.00.
- c) The account balance after 3 years is approximately \$3,500.00.
- d) If the four \$250 deposits are instead treated as a flat \$1,000.00 deposit at year-end, the resulting 4-year account balance is approximately \$4,506.11.
- e) The difference between the correctly computed 4-year balance and the simplified calculation is approximately \$200.00.

Solution – Task 7

Given:

- D (quarterly deposit) = \$250
- r (annual interest rate) = 0.08
- Deposits made at the end of each quarter; interest credited once per year

Formula(s):

- Year-end equivalent of quarterly deposits: $D \cdot (1+r \cdot 3/4) + D \cdot (1+r \cdot 2/4) + D \cdot (1+r \cdot 1/4) + D \cdot (1+r \cdot 0) = D \cdot (4 + 1.5r)$
- Future value after N years, treating each year's equivalent as an ordinary annuity payment: $(\text{year-end value}/r) \cdot [(1+r)^N - 1]$

Step-by-Step Calculations:

Year-end equivalent = $250 \times (4 + 1.5 \times 0.08) = 250 \times 4.12 = \$1,030.00$ (not \$1,100.00).

FV after 4 years = $(1,030.00/0.08) \times [(1.08)^4 - 1] = 12,875 \times 0.360489 \approx \$4,641.30$ (not \$4,700.00).

FV after 3 years = $(1,030.00/0.08) \times [(1.08)^3 - 1] = 12,875 \times 0.259712 \approx \$3,343.79$ (not \$3,500.00).

Simplified (wrong) year-end value ignoring mid-year interest = $4 \times 250 = \$1,000.00$;
simplified FV after 4 years = $(1,000/0.08) \times [(1.08)^4 - 1] \approx \$4,506.11$, which is indeed less than the correct \$4,641.30.

Difference = $4,641.30 - 4,506.11 \approx \135.18 (not \$200.00).

Explanation of Each Statement:

- a) **FALSE.** Applying simple interest to each of the four quarterly deposits for its own remaining fraction of the year and summing the results gives $250 \times (4 + 1.5 \times 0.08) = \$1,030.00$, not \$1,100.00.

- b) FALSE.** Treating \$1,030.00 as an equivalent single annual deposit and compounding it as an ordinary annuity for 4 years gives approximately \$4,641.30, not \$4,700.00.
- c) FALSE.** Using the same year-end equivalent value but compounding for only 3 years instead of 4 gives approximately \$3,343.79, not \$3,500.00.
- d) TRUE.** Ignoring the small simple-interest boost that each quarterly deposit earns before the year-end crediting date understates the true equivalent annual deposit (using \$1,000.00 instead of the correct \$1,030.00), which in turn understates the compounded 4-year balance, giving about \$4,506.11 instead of the true \$4,641.30 — confirming the simplified approach falls short of the correct answer.
- e) FALSE.** The gap between the correct balance ($\approx \$4,641.30$) and the simplified, interest-ignoring balance ($\approx \$4,506.11$) is only about \$135.18, not \$200.00.

Final Answers — a: FALSE b: FALSE c: FALSE d: TRUE e: FALSE

Task 8. Home Mortgage: Outstanding Balance Partway Through the Term

Difficulty: 4.5/5

A family takes out a \$200,000 home mortgage at a nominal annual interest rate of 6%, compounding monthly, to be repaid with equal payments at the end of each month over 20 years. After making exactly 5 years of payments (60 monthly payments), they want to know how much principal is still outstanding.

Statements (True / False):

- a) The required monthly payment is approximately \$1,432.86.
- b) The outstanding balance immediately after the 60th monthly payment is approximately \$169,799.20.
- c) After 5 years of payments, more than 25% of the original \$200,000 principal has been repaid.
- d) The total interest paid during just the first 5 years is approximately \$55,770.92.
- e) The total interest paid over the entire 20-year life of the loan is approximately \$120,000.00.

Solution – Task 8

Given:

- K (mortgage amount) = \$200,000
- Nominal annual rate = 6%, compounding monthly
- Term = 20 years (240 monthly payments)
- m = 60 payments already made

Formula(s):

- Monthly periodic rate: r = nominal annual rate / 12
- Equal-payment formula: $a = rK/[1-(1+r)^{-n}]$
- Outstanding balance after m payments (of an n -payment loan): remaining balance = $(a/r) \cdot [1-(1+r)^{-(n-m)}]$

Step-by-Step Calculations:

$$r = 0.06/12 = 0.005; n = 20 \times 12 = 240 \text{ months.}$$

$$a = 0.005 \times 200,000 / [1 - (1.005)^{-240}] = 1,000 / 0.697884 \approx \$1,432.86.$$

$$\text{Remaining balance after 60 payments} = (1,432.86 / 0.005) \times [1 - (1.005)^{-180}] \approx 286,572.42 \times 0.592652 \approx \$169,799.20.$$

Principal repaid in 5 years = $200,000 - 169,799.20 = \$30,200.80$; as a fraction of the original principal, $30,200.80 / 200,000 \approx 15.10\%$, which is not more than 25%.

$$\text{Total paid in 5 years} = 1,432.86 \times 60 \approx \$85,971.73; \text{ interest paid in 5 years} = 85,971.73 - 30,200.80 \approx \$55,770.92.$$

$$\text{Total paid over all 240 months} = 1,432.86 \times 240 \approx \$343,886.91; \text{ total interest over the full term} = 343,886.91 - 200,000.00 \approx \$143,886.91 \text{ (not } \$120,000.00).$$

Explanation of Each Statement:

- a) **TRUE.** Plugging $r=0.005$, $K=200,000$, $n=240$ into the payment formula gives $a \approx \$1,432.86$, matching exactly.

- b) TRUE.** The remaining 180 monthly payments still owed after month 60 form an ordinary annuity of their own; discounting them back at the same monthly rate gives an outstanding balance of approximately \$169,799.20.
- c) FALSE.** Since the balance has only fallen from \$200,000 to about \$169,799.20 after 5 years, only about \$30,200.80, or roughly 15.1%, of the original principal has actually been repaid so far — nowhere near 25%. Early mortgage payments go disproportionately toward interest rather than principal, which is exactly why the repaid fraction is still so small this early in a 20-year loan.
- d) TRUE.** Subtracting the \$30,200.80 of principal actually repaid from the \$85,971.73 total paid over the first 60 months leaves approximately \$55,770.92 in interest paid during those first 5 years.
- e) FALSE.** Multiplying the monthly payment by all 240 months and subtracting the original \$200,000 principal gives a total lifetime interest cost of approximately \$143,886.91, considerably more than the \$120,000.00 claimed.

Final Answers — a: TRUE b: TRUE c: FALSE d: TRUE e: FALSE

Task 9. Fixed Repayment Schedule on a Large Equipment Loan

Difficulty: 4.75/5

A manufacturing company borrows \$120,000 at 14% annual interest and chooses to repay a fixed \$25,000 at the end of each year (rather than solving for the exact equal-payment amount), continuing until the loan is retired, with a final smaller payment to clear whatever balance remains.

Statements (True / False):

- a) The smallest whole number of yearly payments needed to retire the loan is $n = 9$.
- b) The ninth and final payment is approximately \$13,100.16.
- c) The total interest paid over the life of the loan is approximately \$105,000.00.
- d) The total amount actually paid over the life of the loan is approximately \$210,000.00.
- e) Assuming all 9 payments are full \$25,000 payments overstates the true total amount paid by more than \$10,000.

Solution – Task 9

Given:

- K (loan amount) = \$120,000
- r (annual interest rate) = 0.14
- a (fixed annual payment) = \$25,000

Formula(s):

- Number of periods needed: smallest integer n with $n \geq [\ln a - \ln(a-rK)]/\ln(1+r)$
- Future value of loan after m years: $K \cdot (1+r)^m$
- Future value of m fixed payments (valued at the time of the m -th payment): $(a/r) \cdot [(1+r)^m - 1]$

Step-by-Step Calculations:

$[\ln(25,000) - \ln(25,000 - 0.14 \times 120,000)]/\ln(1.14) = [\ln(25,000) - \ln(8,200)]/\ln(1.14) \approx 8.508$, so the smallest integer n satisfying the condition is $n = 9$ (8 full payments plus a final 9th payment).

Future value of the loan after 8 years = $120,000 \times (1.14)^8 \approx \$342,310.37$.

Future value of the 8 payments of \$25,000, valued at year 8 = $(25,000/0.14) \times [(1.14)^8 - 1] \approx \$330,819.00$.

Remaining debt at year 8 = $342,310.37 - 330,819.00 \approx \$11,491.37$; carrying forward one more year: $11,491.37 \times 1.14 \approx \$13,100.16$.

Total actually paid = $8 \times 25,000 + 13,100.16 = 200,000.00 + 13,100.16 \approx \$213,100.16$; true total interest = $213,100.16 - 120,000.00 \approx \$93,100.16$ (not \$105,000.00).

If all 9 payments were mistakenly treated as full \$25,000 payments, the (incorrect) total would be $9 \times 25,000 = \$225,000.00$, overstating the true total of \$213,100.16 by $225,000.00 - 213,100.16 \approx \$11,899.84$, which is indeed more than \$10,000.

Explanation of Each Statement:

- a) **TRUE.** Evaluating $[\ln a - \ln(a-rK)]/\ln(1+r)$ with $a=25,000$, $r=0.14$, $K=120,000$ gives approximately 8.508; rounding up to the next whole number gives $n=9$ total payments needed.
- b) **TRUE.** Comparing the loan's accumulated value after 8 years to the accumulated value of the eight \$25,000 payments leaves a small remaining balance, which grows one more year of interest into a final payment of approximately \$13,100.16.
- c) **FALSE.** Since the final, ninth payment is smaller than \$25,000 (only about \$13,100.16), simply multiplying \$25,000 by 9 and subtracting \$120,000 overstates the true interest cost; the actual total interest paid is closer to \$93,100.16, not \$105,000.00.
- d) **FALSE.** Adding the eight full \$25,000 payments (\$200,000.00) to the smaller final payment (about \$13,100.16) gives a true total of approximately \$213,100.16, not \$210,000.00.
- e) **TRUE.** Treating the final payment as if it were also a full \$25,000 payment produces a total of \$225,000.00, which exceeds the true total of about \$213,100.16 by roughly \$11,899.84 — an overstatement of more than \$10,000, confirming the claim.

Final Answers — a: TRUE b: TRUE c: FALSE d: FALSE e: TRUE

Task 10. Capstone: Equipment Loan Paid as an Annuity Due, Plus a Separate Reserve Fund

Difficulty: 5/5

A manufacturing company borrows \$90,000 to buy new equipment, to be repaid in 8 equal annual instalments with the first payment due immediately (an annuity due) and the rest due at the beginning of each following year, at an annual interest rate of 12%. Independently, to prepare for future maintenance costs, the company also deposits \$300 at the end of each quarter into a separate reserve account that pays 9% annual interest, credited once per year (with simple interest applied to each quarterly deposit for its remaining fraction of the year, exactly as in a within-period deposit calculation).

Statements (True / False):

- a) The required annuity-due payment on the equipment loan is approximately \$16,176.12.
- b) The interest portion of the second loan payment is approximately \$8,858.87.
- c) The interest portion of the third loan payment is larger than the interest portion of the second payment.
- d) The year-end equivalent value of the reserve fund's quarterly deposits made during a single year is \$1,240.50, and the reserve fund's balance after 3 years is approximately \$4,066.48.
- e) The combined total of the loan's first three payments is less than the reserve fund's balance after 3 years.

Solution – Task 10

Given:

- Equipment loan: $K = \$90,000$, $r = 0.12$, $n = 8$ payments, first one immediate
- Reserve fund: quarterly deposit = \$300, annual rate = 0.09, interest credited once per year, $N = 3$ years

Formula(s):

- Annuity-due present value: $K = a + (a/r) \cdot [1 - (1+r)^{-(n-1)}]$
- Amortization after the immediate first payment: remaining loan behaves as an ordinary annuity of a for the remaining $n-1$ payments; interest each period = $r \times$ balance at start of that period
- Year-end equivalent of quarterly deposits: $D \cdot (4 + 1.5r)$; future value after N years = $(\text{year-end value}/r) \cdot [(1+r)^N - 1]$

Step-by-Step Calculations:

Combined factor = $1 + (1/0.12) \times [1 - (1.12)^{-7}] \approx 1 + 4.563757 = 5.563757$; $a = 90,000/5.563757 \approx \$16,176.12$.

Balance immediately after the first (immediate) payment = $90,000 - 16,176.12 = \$73,823.88$.

Second payment (1 year later): interest = $0.12 \times 73,823.88 \approx \$8,858.87$; principal = $16,176.12 - 8,858.87 \approx \$7,317.26$; new balance $\approx \$66,506.62$.

Third payment (2 years later): interest = $0.12 \times 66,506.62 \approx \$7,980.79$, which is smaller than the second payment's interest of \$8,858.87, not larger; principal $\approx \$8,195.33$; new balance $\approx \$58,311.30$.

Reserve fund year-end equivalent = $300 \times (4 + 1.5 \times 0.09) = 300 \times 4.135 = \$1,240.50$; FV after 3 years = $(1,240.50/0.09) \times [(1.09)^3 - 1] \approx 13,783.33 \times 0.295029 \approx \$4,066.48$.

Total of the loan's first three payments = $3 \times 16,176.12 \approx \$48,528.36$, which is far more than the reserve fund's 3-year balance of about \$4,066.48, not less.

Explanation of Each Statement:

- a) TRUE.** Because the first payment is made immediately and only the remaining 7 payments are discounted as an ordinary annuity, the combined present-value factor works out to approximately 5.563757, and dividing \$90,000 by that factor gives a $\approx \$16,176.12$.
- b) TRUE.** After the immediate first payment reduces the balance to \$73,823.88, that balance accrues a full year of 12% interest before the second payment is made, giving an interest portion of approximately \$8,858.87 for that second payment.
- c) FALSE.** By the time of the third payment, the outstanding balance has already shrunk (to about \$66,506.62), so the interest charged on it (about \$7,980.79) is smaller than the interest charged in the previous period (about \$8,858.87), not larger — interest portions shrink from payment to payment as the balance amortizes down.
- d) TRUE.** Applying simple interest to each quarterly \$300 deposit for its own remaining fraction of the year gives a year-end equivalent of \$1,240.50, and compounding that equivalent amount annually as an ordinary annuity for 3 years gives a reserve balance of approximately \$4,066.48.
- e) FALSE.** Three annual loan payments of about \$16,176.12 each add up to roughly \$48,528.36, which vastly exceeds the reserve fund's much smaller 3-year balance of about \$4,066.48 — the loan payments are far larger than the reserve fund total, not smaller, since the loan involves a much larger principal amount than the modest quarterly reserve deposits.

Final Answers — a: TRUE b: TRUE c: FALSE d: TRUE e: FALSE

Task 11. Capstone: Comparing Three Payment Schedules for a Building Site

Difficulty: 5/5

A construction firm wants to buy a building site and is choosing among three payment schedules. Schedule I: pay \$500,000 in cash immediately. Schedule II: pay \$95,000 per year for 7 years, with the first instalment paid immediately (an annuity due). Schedule III: pay \$150,000 in cash immediately, plus \$60,000 per year for 10 years, with the first of these instalments paid one year later (an ordinary annuity). The firm wants to know which schedule is cheapest in present-value terms first at a 9% annual interest rate, and then at a 13% annual interest rate.

Statements (True / False):

- a) At a 9% annual rate, the present value of Schedule II is approximately \$521,162.27.
- b) At a 9% annual rate, the present value of Schedule III is approximately \$540,000.00.
- c) At a 9% annual rate, Schedule I is the cheapest of the three schedules.
- d) At a 13% annual rate, the present value of Schedule II is approximately \$474,767.23.
- e) At a 13% annual rate, Schedule III becomes the cheapest of the three schedules.

Solution – Task 11

Given:

- Schedule I: \$500,000 cash today
- Schedule II: $a = \$95,000/\text{year}$, $n = 7$, first payment immediate
- Schedule III: \$150,000 cash today + $a = \$60,000/\text{year}$, $n = 10$, first of these payments one year later
- Interest rates considered: 9% and 13% per year

Formula(s):

- Annuity-due present value: $PV = a + (a/r) \cdot [1 - (1+r)^{-(n-1)}]$
- Ordinary-annuity present value: $PV = (a/r) \cdot [1 - (1+r)^{-n}]$
- Schedule III present value: upfront cash + ordinary-annuity present value of the later instalments

Step-by-Step Calculations:

At $r=0.09$: $PV_{II} = 95,000 + (95,000/0.09) \times [1 - (1.09)^{-6}] \approx 95,000 + 1,055,556 \times 0.403471 \approx \$521,162.27$.

At $r=0.09$: $PV_{III} = 150,000 + (60,000/0.09) \times [1 - (1.09)^{-10}] \approx 150,000 + 666,667 \times 0.577589 \approx \$535,059.46$ (not \$540,000.00).

Comparing at 9%: $PV_I = \$500,000.00 < PV_{II} \approx \$521,162.27 < PV_{III} \approx \$535,059.46$, so Schedule I is cheapest.

At $r=0.13$: $PV_{II} = 95,000 + (95,000/0.13) \times [1 - (1.13)^{-6}] \approx 95,000 + 730,769 \times 0.518945 \approx \$474,767.23$.

At $r=0.13$: $PV_{III} = 150,000 + (60,000/0.13) \times [1 - (1.13)^{-10}] \approx 150,000 + 461,538 \times 0.705115 \approx \$475,574.61$.

Comparing at 13%: $PV_{II} \approx \$474,767.23 < PV_{III} \approx \$475,574.61 < PV_I = \$500,000.00$, so Schedule II is now cheapest, not Schedule III – though Schedule III has come very

close to overtaking it.

Explanation of Each Statement:

- a) **TRUE.** Applying the annuity-due formula with $a=95,000$, $r=0.09$, $n=7$ gives $PV_{II} \approx \$521,162.27$, matching exactly.
- b) **FALSE.** Adding the \$150,000 upfront cash to the ordinary-annuity present value of the later \$60,000 payments (discounted at 9% for 10 years) gives approximately \$535,059.46, not \$540,000.00.
- c) **TRUE.** At 9%, the three present values rank as \$500,000.00 (Schedule I) < \$521,162.27 (Schedule II) < \$535,059.46 (Schedule III), so paying cash outright is the cheapest option at this interest rate.
- d) **TRUE.** Recomputing Schedule II's present value at the higher rate of 13% gives $PV_{II} \approx \$474,767.23$, matching exactly.
- e) **FALSE.** At 13%, Schedule III's present value (about \$475,574.61) is still slightly higher than Schedule II's (about \$474,767.23) — Schedule II, not Schedule III, is the cheapest schedule at this higher rate, even though Schedule III has narrowed the gap considerably compared to how far behind it was at 9%.

Final Answers — a: TRUE b: FALSE c: TRUE d: TRUE e: FALSE

Task 12. Capstone: Comparing Three Payment Schedules for a Hospital Imaging Center

Difficulty: 5/5

A hospital system is negotiating the purchase of a new imaging center and equipment package, and is choosing among three payment schedules. Schedule I: pay \$850,000 in cash immediately. Schedule II: pay \$140,000 per year for 9 years, with the first instalment paid immediately (an annuity due). Schedule III: pay \$300,000 in cash immediately, plus \$80,000 per year for 11 years, with the first of these instalments paid one year later (an ordinary annuity). The hospital wants to know which schedule is cheapest in present-value terms first at an 8% annual interest rate, and then at a 12% annual interest rate.

Statements (True / False):

- a) At an 8% annual rate, the present value of Schedule II is approximately \$944,529.45.
- b) At an 8% annual rate, the present value of Schedule III is approximately \$871,117.14.
- c) At an 8% annual rate, Schedule I is the cheapest of the three schedules.
- d) At a 12% annual rate, the present value of Schedule III is approximately \$775,015.93.
- e) At a 12% annual rate, Schedule II becomes the cheapest of the three schedules.

Solution – Task 12

Given:

- Schedule I: \$850,000 cash today
- Schedule II: $a = \$140,000/\text{year}$, $n = 9$, first payment immediate
- Schedule III: \$300,000 cash today + $a = \$80,000/\text{year}$, $n = 11$, first of these payments one year later
- Interest rates considered: 8% and 12% per year

Formula(s):

- Annuity-due present value: $PV = a + (a/r) \cdot [1 - (1+r)^{-(n-1)}]$
- Ordinary-annuity present value: $PV = (a/r) \cdot [1 - (1+r)^{-n}]$
- Schedule III present value: upfront cash + ordinary-annuity present value of the later instalments

Step-by-Step Calculations:

At $r=0.08$: $PV_{II} = 140,000 + (140,000/0.08) \times [1 - (1.08)^{-8}] \approx 140,000 + 1,750,000 \times 0.459640 \approx \$944,529.45$.

At $r=0.08$: $PV_{III} = 300,000 + (80,000/0.08) \times [1 - (1.08)^{-11}] \approx 300,000 + 1,000,000 \times 0.571146 \approx \$871,117.14$.

Comparing at 8%: $PV_I = \$850,000.00 < PV_{III} \approx \$871,117.14 < PV_{II} \approx \$944,529.45$, so Schedule I is cheapest.

At $r=0.12$: $PV_{III} = 300,000 + (80,000/0.12) \times [1 - (1.12)^{-11}] \approx 300,000 + 666,667 \times 0.712524 \approx \$775,015.93$.

At $r=0.12$: $PV_{II} = 140,000 + (140,000/0.12) \times [1 - (1.12)^{-8}] \approx 140,000 + 1,166,667 \times 0.596117 \approx \$835,469.57$.

Comparing at 12%: $PV_{III} \approx \$775,015.93 < PV_{II} \approx \$835,469.57 < PV_I = \$850,000.00$, so Schedule III becomes cheapest, not Schedule II.

Explanation of Each Statement:

- a) **TRUE.** Applying the annuity-due formula with $a=140,000$, $r=0.08$, $n=9$ gives $PV_{II} \approx \$944,529.45$, matching exactly.
- b) **TRUE.** Adding the \$300,000 upfront cash to the ordinary-annuity present value of the later \$80,000 payments (discounted at 8% for 11 years) gives approximately \$871,117.14, matching exactly.
- c) **TRUE.** At 8%, the three present values rank as \$850,000.00 (Schedule I) < \$871,117.14 (Schedule III) < \$944,529.45 (Schedule II), so paying cash outright is the cheapest option at this interest rate.
- d) **TRUE.** Recomputing Schedule III's present value at the higher rate of 12% gives $PV_{III} \approx \$775,015.93$, matching exactly.
- e) **FALSE.** At 12%, Schedule III's present value (about \$775,015.93) is actually lower than Schedule II's (about \$835,469.57) — Schedule III, not Schedule II, becomes the cheapest option at this higher rate, since raising the discount rate shrinks the present value of Schedule III's smaller, longer-running instalments even more than it shrinks Schedule II's larger, shorter-running ones.

Final Answers — a: TRUE b: TRUE c: TRUE d: TRUE e: FALSE

Task 13. Capstone: Comparing Three Payment Schedules for a Cargo Vessel

Difficulty: 5/5

A shipping company is negotiating the purchase of a cargo vessel and is choosing among three payment schedules. Schedule I: pay \$2,400,000 in cash immediately. Schedule II: pay \$340,000 per year for 10 years, with the first instalment paid immediately (an annuity due). Schedule III: pay \$600,000 in cash immediately, plus \$250,000 per year for 9 years, with the first of these instalments paid one year later (an ordinary annuity). The company wants to know how the ranking of Schedule I versus Schedule II changes between a 7.5% annual interest rate and a 11.5% annual interest rate.

Statements (True / False):

- a) At a 7.5% annual rate, the present value of Schedule II is approximately \$2,508,821.59.
- b) At a 7.5% annual rate, the present value of Schedule III is approximately \$2,250,000.00.
- c) At a 7.5% annual rate, Schedule I is cheaper than Schedule II.
- d) At an 11.5% annual rate, the present value of Schedule II is approximately \$2,100,000.00.
- e) At an 11.5% annual rate, Schedule I is still cheaper than Schedule II.

Solution – Task 13

Given:

- Schedule I: \$2,400,000 cash today
- Schedule II: $a = \$340,000/\text{year}$, $n = 10$, first payment immediate
- Schedule III: \$600,000 cash today + $a = \$250,000/\text{year}$, $n = 9$, first of these payments one year later
- Interest rates considered: 7.5% and 11.5% per year

Formula(s):

- Annuity-due present value: $PV = a + (a/r) \cdot [1 - (1+r)^{-(n-1)}]$
- Ordinary-annuity present value: $PV = (a/r) \cdot [1 - (1+r)^{-n}]$
- Schedule III present value: upfront cash + ordinary-annuity present value of the later instalments

Step-by-Step Calculations:

At $r=0.075$: $PV_{II} = 340,000 + (340,000/0.075) \times [1 - (1.075)^{-9}] \approx 340,000 + 4,533,333 \times 0.478550 \approx \$2,508,821.59$.

At $r=0.075$: $PV_{III} = 600,000 + (250,000/0.075) \times [1 - (1.075)^{-9}] \approx 600,000 + 3,333,333 \times 0.478550 \approx \$2,194,721.76$ (not \$2,250,000.00).

Comparing at 7.5%: $PV_I = \$2,400,000.00 < PV_{II} \approx \$2,508,821.59$, so cash is cheaper than the annuity-due schedule at this rate.

At $r=0.115$: $PV_{II} = 340,000 + (340,000/0.115) \times [1 - (1.115)^{-9}] \approx 340,000 + 2,956,522 \times 0.624582 \approx \$2,186,561.89$ (not \$2,100,000.00).

At $r=0.115$: $PV_{III} \approx 600,000 + (250,000/0.115) \times [1 - (1.115)^{-9}] \approx \$1,957,766.09$.

Comparing at 11.5%: $PV_{II} \approx \$2,186,561.89 < PV_I = \$2,400,000.00$, so the annuity-due schedule has now become cheaper than paying cash – the ranking between Schedules I and II has flipped compared to the 7.5% case (Schedule III remains the cheapest of

all three at both rates).

Explanation of Each Statement:

a) TRUE. Applying the annuity-due formula with $a=340,000$, $r=0.075$, $n=10$ gives $PV_{\text{II}} \approx \$2,508,821.59$, matching exactly.

b) FALSE. Adding the \$600,000 upfront cash to the ordinary-annuity present value of the later \$250,000 payments (discounted at 7.5% for 9 years) gives approximately \$2,194,721.76, not \$2,250,000.00.

c) TRUE. At 7.5%, \$2,400,000.00 (Schedule I) is indeed less than approximately \$2,508,821.59 (Schedule II), so paying cash outright beats the annuity-due schedule at this lower rate.

d) FALSE. Recomputing Schedule II's present value at the higher rate of 11.5% gives $PV_{\text{II}} \approx \$2,186,561.89$, not \$2,100,000.00.

e) FALSE. At the higher rate of 11.5%, Schedule II's present value drops to approximately \$2,186,561.89, which is now lower than the \$2,400,000.00 cash price — the annuity-due schedule has overtaken cash as the cheaper option, reversing the ranking that held at 7.5%. Raising the discount rate reduces the present value of a stream of future payments more sharply than it affects an already-immediate cash amount, which is exactly what causes this kind of ranking reversal.

Final Answers — a: TRUE b: FALSE c: TRUE d: FALSE e: FALSE

This worksheet relies exclusively on the equal-payment loan formula, the interest/principal amortization breakdown, the annuity-due adjustment for an immediate first payment, the method for finding the number of periods needed under a fixed payment amount (with a smaller final payment), and the simple-interest treatment of deposits made within a single interest-crediting period.